Arduino Pin Usage & Functions Worksheet

Name: _____ **Date:** _____

Part 1: Matching Functions

Draw a line or write the letter of the correct description next to each Arduino function.

D `digitalWrite()`

B `digitalRead()`

A `analogRead()`

C `analogWrite()`

A. Reads a continuous voltage between 0V and 5V and converts it to a number from 0 to 1023. Can only be used on specific pins.
B. Reads whether a pin is set to HIGH (approx. $\geq$ 2.5V) or LOW (approx. $<$ 2.5V). Can be used on any pin.
C. Uses Pulse Width Modulation (PWM) to simulate an analog value by rapidly turning the pin on and off. Takes a value from 0 to 255.
D. Turns a pin completely ON (5V) or completely OFF (0V). Can be used on any pin.

Part 2: Fill in the Blanks (PWM and Tone)

1. When using `analogWrite()`, you are providing a value between 0 and 255. A value of **0** means the duty cycle is 0 %, while a value of **127** means the pin is HIGH for approximately 50 % of the time.

2. True or False: The `analogWrite()` function works by actually lowering the voltage output of the pin to a steady value somewhere between 0V and 5V. False

3. The `tone()` function is used to output a specific frequency. Unlike `analogWrite()`, which uses native PWM, `tone()` generates its frequency by utilizing ~~the~~ one of Arduino's built-in timers.

Part 3: Pin Selection Scenarios

Based on the lecture caveats, choose the best pin(s) or pin ranges for each scenario. If there are pins you should specifically avoid, mention them.

4. **Scenario:** You want to read the variable voltage from a temperature sensor that outputs between 0V and 5V.
   - **Which group of pins MUST you use?** A0 - A15

5. **Scenario:** You want to dim an LED so it appears to be at 25% brightness.
   - **Which function do you need?** analogWrite
   - **What is a safe pin number you could use for this?** D2-15 (not 13) D44-46 ✓
   - **Which PWM pin should you generally avoid using because it's attached to the board's built-in LED?** D13

6. **Scenario:** You just need to turn a water pump completely ON or completely OFF.
   - **Can you use an Analog pin (like A3) for this if you run out of digital pins?** yes (but don't)
   - **Why should you avoid using Digital Pins 0 and 1 for this pump?** used for laptop communication

Part 4: Short Answer

7. Your classmate wrote the following code to read a simple on/off push-button:
   `int buttonState = analogRead(4);`
   Explain to your classmate why this is the wrong function to use for a simple on/off button, and tell them which function they should use instead.

   Two answers: analogRead ✗ on digital pin 4
   digitalRead makes more sense

Answer Key (For Instructor Reference)

Part 1

1. D
2. B
3. A
4. C

Part 2

5. 0%, 50%
6. False (It pulses 5V and 0V rapidly; it does not output a steady middle voltage).
7. Timers (specifically Timer 2).

Part 3

8. Analog-to-digital pins (A0 - A15).
9. `analogWrite()`; Any valid PWM pin (e.g., D2-D12, D44-D46); Avoid D13.
10. Yes (though not recommended so you don't get confused); Pins 0 and 1 overlap with communication/Serial monitor tech.

Part 4

11. A simple push-button only has two states (on or off / HIGH or LOW). `analogRead()` is for reading a range of voltages (0-1023) on analog pins. They should use `digitalRead()` instead, which simply checks if the voltage is HIGH ($\geq$ 2.5V) or LOW, and it can be used on any digital pin.